Gen2sat:
A Generic Tool for Reasoning with Non-classical Logics

Yoni Zohar – Tel Aviv University

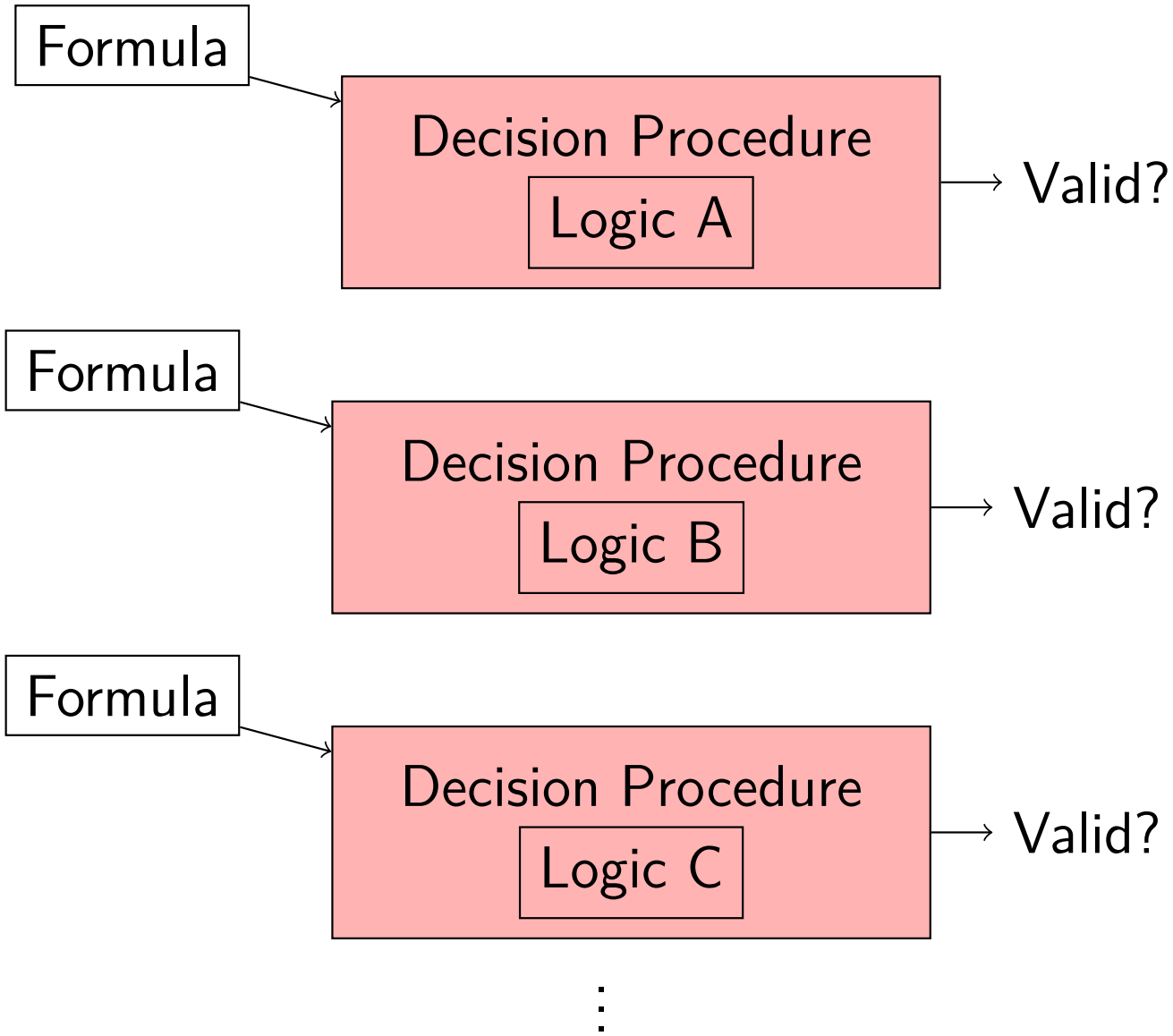
Joint work with Ori Lahav and Anna Zamansky

AIC @ SRI
August 24, 2017

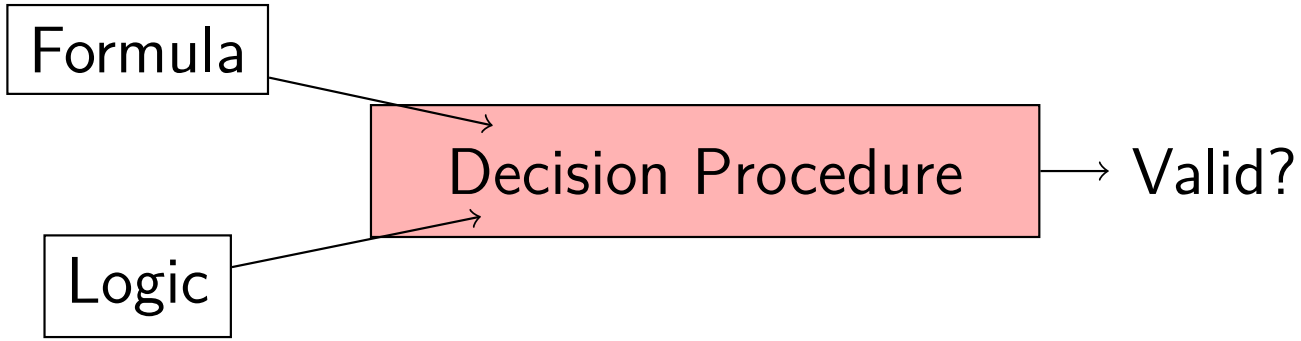
Motivation

- Propositional classical logic:
  - A lot of research
  - Used in applications
- Propositional Non-classical logics:
  - Many-valued and fuzzy logics: $v(A) = 0.5$
  - Paraconsistent logics: $A, \neg A \nvdash B$
  - Intuitionistic logic: $\nvdash A \vee \neg A$
  - Modal logics: $\Box A$
  - A lot of research
  - Few are used in applications
- A possible explanation:
  - Lack of available tools for reasoning with non-classical logics
  - One has to develop a reasoning tool from scratch for each logic

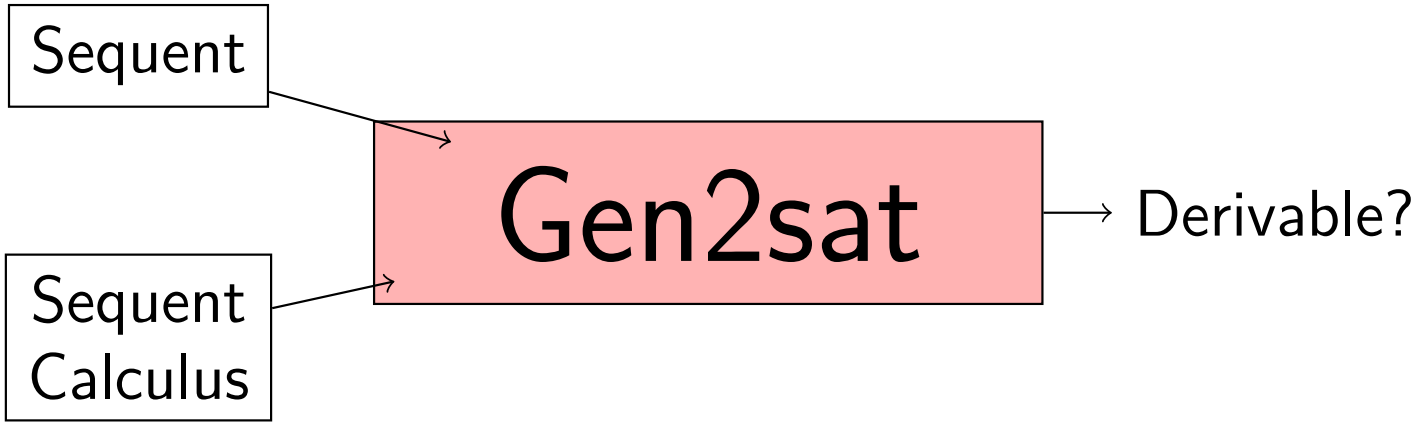
Tools for Non-classical Logics



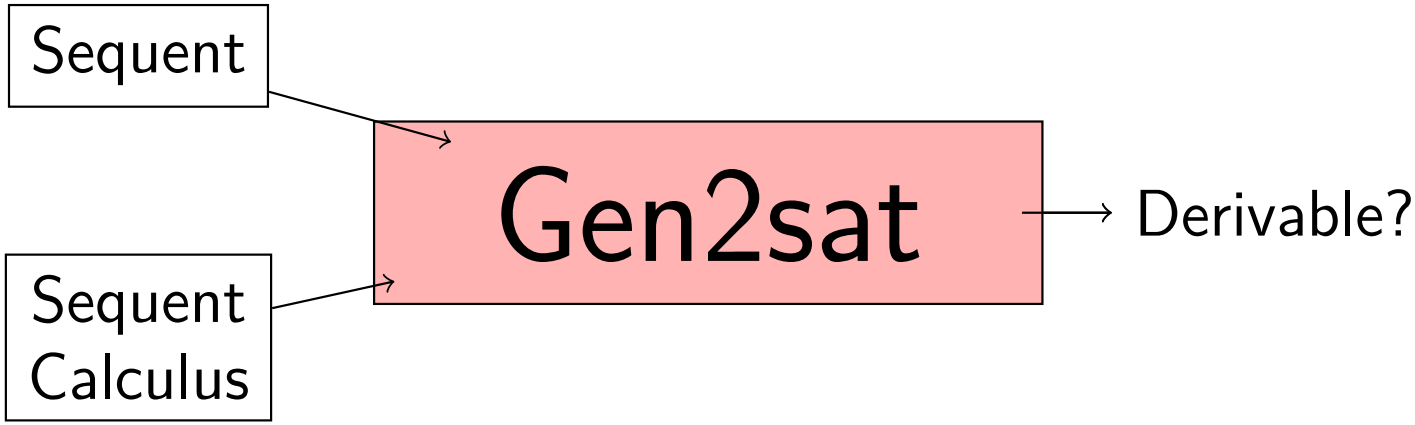
Tool for Non-classical Logics



Tool for Non-classical Logics

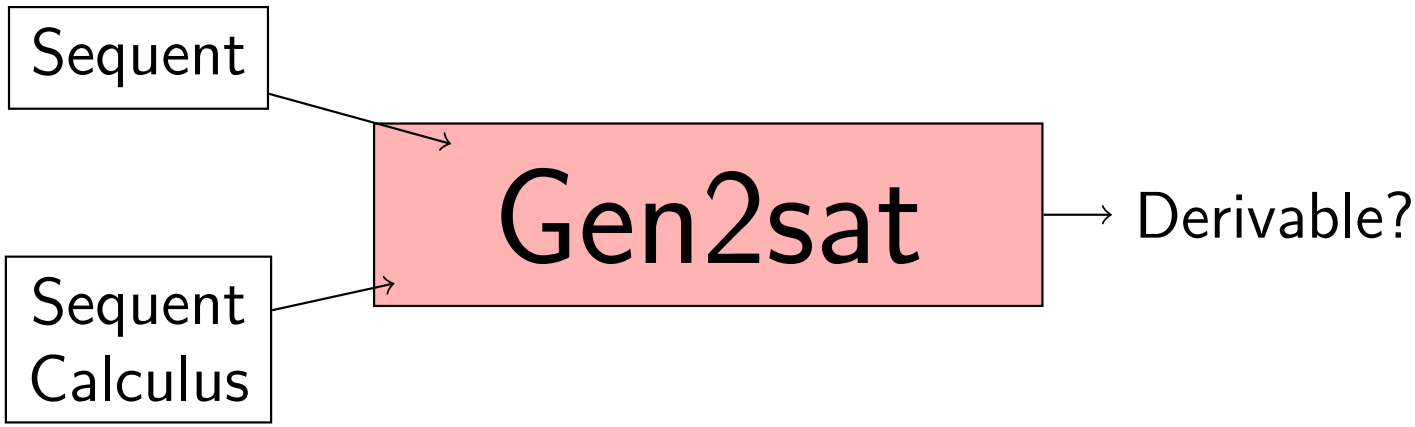


Tool for Non-classical Logics





Tool for Non-classical Logics



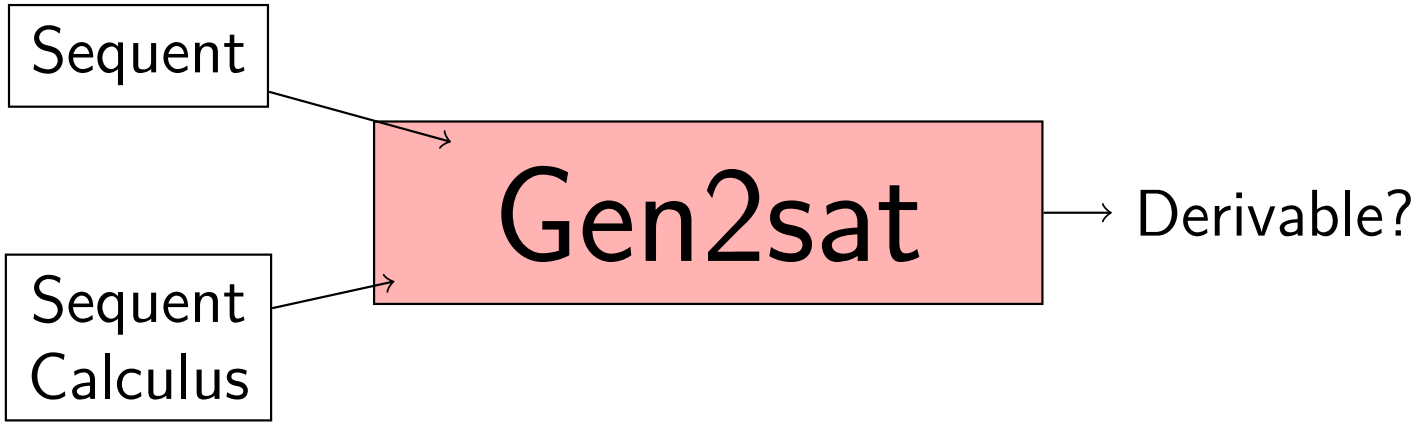
Works for Propositional Pure Analytic Sequent Calculi with "Next" Operators

IN:
classical logic
3-valued logics
4-valued logics
paraconsistent logics
...

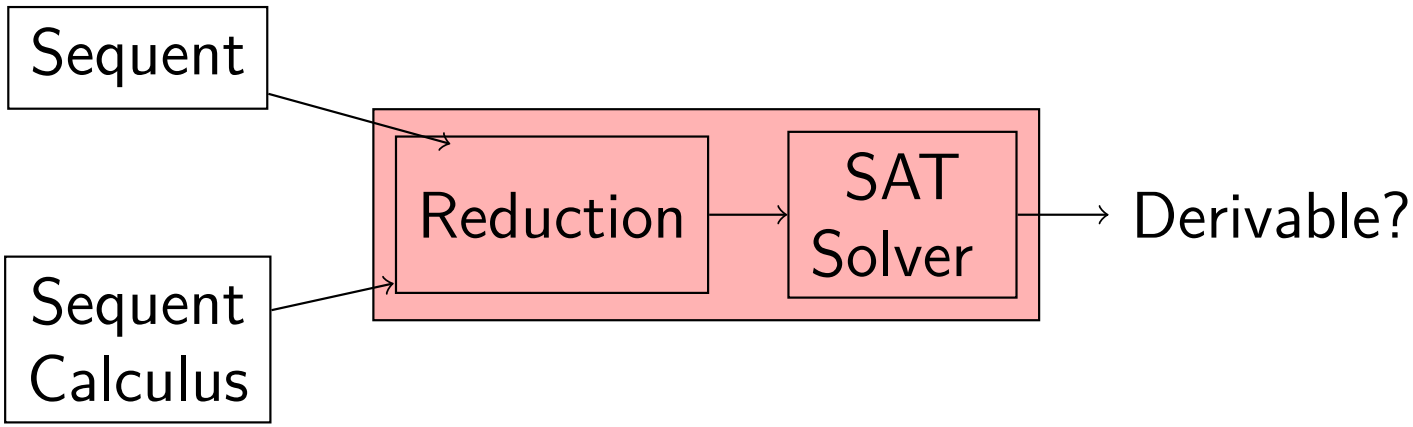


OUT:
intuitionistic logic
relevance logics
fuzzy logics
first-order logics
...

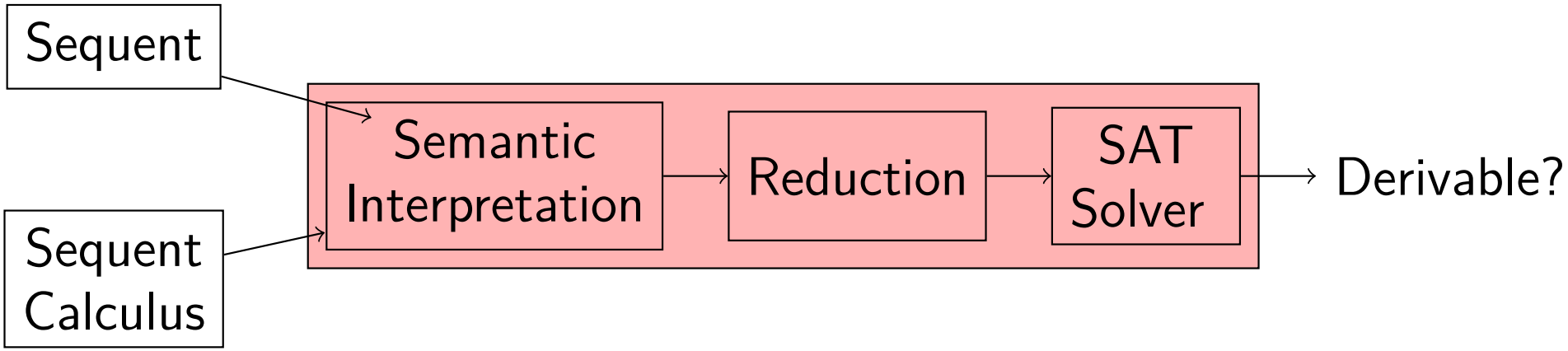
Tool for Non-classical Logics



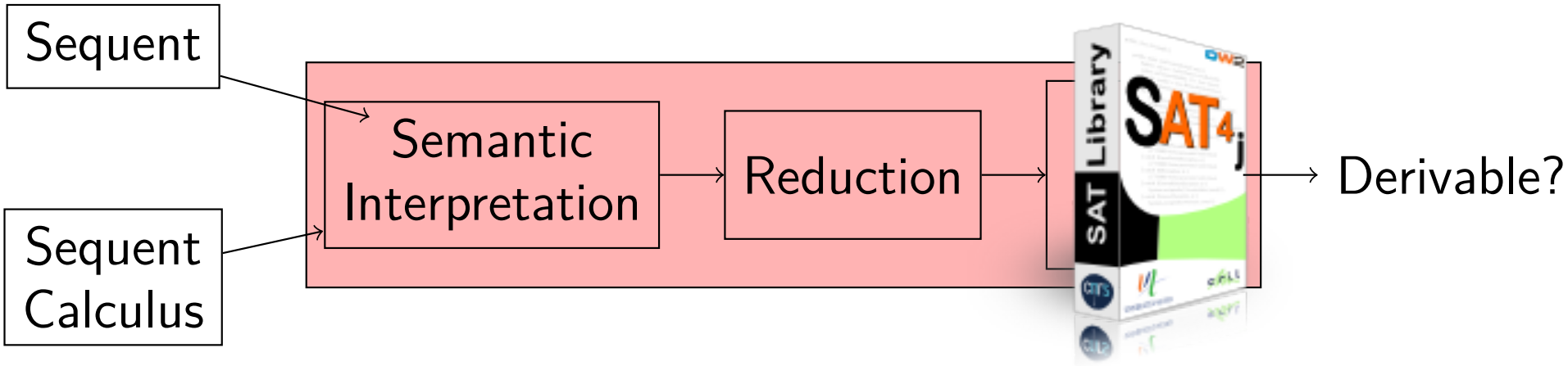
Tool for Non-classical Logics



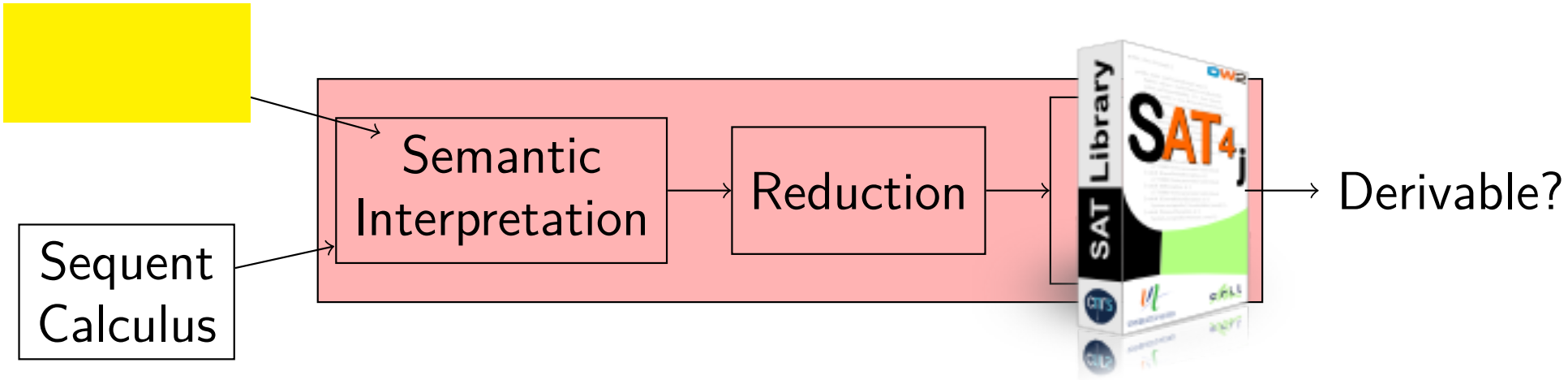
Tool for Non-classical Logics



Tool for Non-classical Logics



Tool for Non-classical Logics



What Are Sequents?

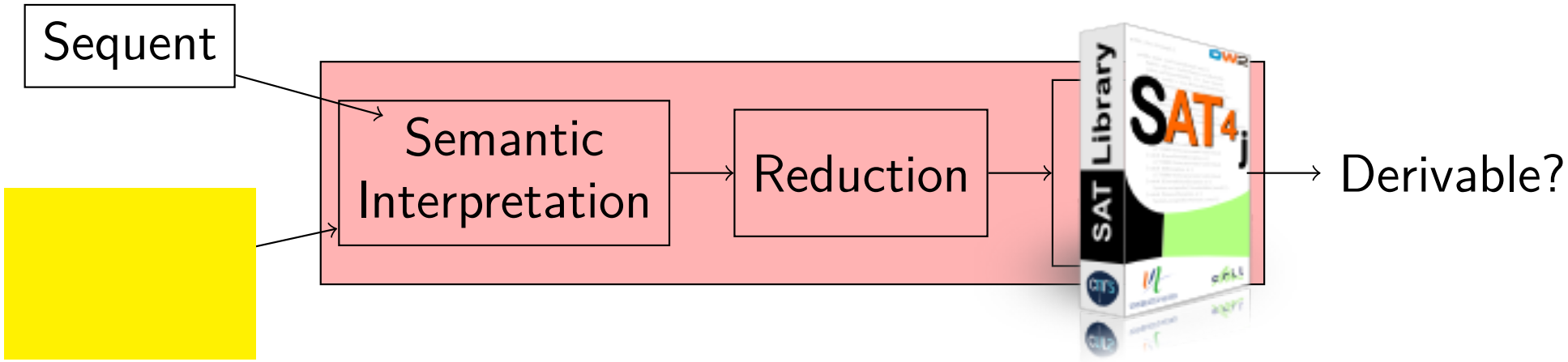
- *Sequents* have the form $\Gamma \Rightarrow \Delta$, where Γ, Δ are finite sets of formulas.
- Intuition:

$$A_1, \ldots, A_n \Rightarrow B_1, \ldots, B_m \quad \rightsquigarrow \quad A_1 \wedge \ldots \wedge A_n \rightarrow B_1 \vee \ldots \vee B_m$$

Example

- $A, B \Rightarrow A \wedge B$
- $A \Rightarrow A \vee B$
- $\Rightarrow A \vee \neg A$
- $A, \neg A \Rightarrow$
- $\Rightarrow A, \neg A$
- $A \Rightarrow A, B, C$

Tool for Non-classical Logics



Sequent Calculi

- Proof systems that manipulate sequents
- Sequent Calculus = finite set of sequent derivation rules

$$\frac{\Gamma_1 \Rightarrow \Delta_1, \ldots, \Gamma_n \Rightarrow \Delta_n}{\Gamma_0 \Rightarrow \Delta_0}$$

Example (Thinking of a Sequent Rule)

$$\frac{\Gamma \Rightarrow A \qquad \Gamma \Rightarrow B}{\Gamma \Rightarrow A \wedge B} \qquad \frac{A \Rightarrow \Delta}{A \wedge B \Rightarrow \Delta} \qquad \frac{B \Rightarrow \Delta}{A \wedge B \Rightarrow \Delta}$$

Sequent Calculi

- Proof systems that manipulate sequents
- Sequent Calculus = finite set of sequent derivation rules

$$\frac{\Gamma_1 \Rightarrow \Delta_1, \ldots, \Gamma_n \Rightarrow \Delta_n}{\Gamma_0 \Rightarrow \Delta_0}$$

Example (Thinking of a Sequent Rule)

$$\frac{\Gamma \Rightarrow A, \Delta \qquad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta} \qquad \frac{\Gamma, A \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad \frac{\Gamma, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta}$$

Examples of Sequent Calculi

The Propositional Fragment of **LK** [Gentzen 1934]

Structural Rules:

$$(id)\ \frac{}{\Gamma, A \Rightarrow A, \Delta} \qquad (cut)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \Delta} \qquad (weak)\ \frac{\Gamma \Rightarrow \Delta}{\Gamma, \Gamma' \Rightarrow \Delta, \Delta'}$$

Logical Rules:

$$(\neg \Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta}{\Gamma, \neg A \Rightarrow \Delta} \qquad (\Rightarrow \neg)\ \frac{\Gamma, A \Rightarrow \Delta}{\Gamma \Rightarrow \neg A, \Delta}$$

$$(\wedge \Rightarrow)\ \frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad (\Rightarrow \wedge)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}$$

$$(\vee \Rightarrow)\ \frac{\Gamma, A \Rightarrow \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \vee B \Rightarrow \Delta} \qquad (\Rightarrow \vee)\ \frac{\Gamma \Rightarrow A, B, \Delta}{\Gamma \Rightarrow A \vee B, \Delta}$$

$$(\supset \Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \supset B \Rightarrow \Delta} \qquad (\Rightarrow \supset)\ \frac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \supset B, \Delta}$$

Analyticity ("subformula property")

Only subformulas of the proved sequent are used

Examples of Sequent Calculi

Łukasiewicz 3-valued Logic [Avron '03]

A sequent calculus for $Ł_3$ is obtained by augmenting the positive fragment of **LK** with some pure rules for negation. For example:

$$(\neg \supset \Rightarrow) \qquad \frac{\Gamma, A, \neg B \Rightarrow \Delta}{\Gamma, \neg(A \supset B) \Rightarrow \Delta}$$

$$(\Rightarrow \neg \supset) \qquad \frac{\Gamma \Rightarrow A, \Delta \qquad \Gamma \Rightarrow \neg B, \Delta}{\Gamma \Rightarrow \neg(A \supset B), \Delta}$$

¬-Analyticity

Only subformulas of the proved sequent and their negations are used

Examples of Sequent Calculi

Calculus for Primal Infon Logic [Gurevich,Neeman '09]

$$(\wedge \Rightarrow)\ \frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \qquad (\Rightarrow \wedge)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow A \wedge B, \Delta}$$

$$(\vee \Rightarrow)\ \frac{\Gamma, A \Rightarrow \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \vee B \Rightarrow \Delta} \qquad (\Rightarrow \vee)\ \frac{\Gamma \Rightarrow A, B, \Delta}{\Gamma \Rightarrow A \vee B, \Delta}$$

$$(\supset \Rightarrow)\ \frac{\Gamma \Rightarrow A, \Delta \quad \Gamma, B \Rightarrow \Delta}{\Gamma, A \supset B \Rightarrow \Delta} \qquad (\Rightarrow \supset)\ \frac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \supset B, \Delta}$$

$$\frac{\Gamma \Rightarrow \Delta}{q\ said\ \Gamma \Rightarrow q\ said\ \Delta}$$ for every principal q

- An extremely efficient propositional logic.
- One of the main logical engines behind MSR DKAL

Analyticity

Only subformulas of the proved sequent are used

Scope of Gen2sat

Pure Analytic sequent calculus with "Next" Operators

- Propositional and structural
- Include pure logical rules that allow any Γ and Δ:

$$\checkmark \ \frac{\Gamma, A \Rightarrow B, \Delta}{\Gamma \Rightarrow A \supset B, \Delta} \qquad \text{X} \ \frac{\Gamma, A \Rightarrow B}{\Gamma \Rightarrow A \supset B}$$

- May include impure rules of the form: $\dfrac{\Gamma \Rightarrow \Delta}{*\Gamma \Rightarrow *\Delta}$
- Analytic (subformula property and its variants)

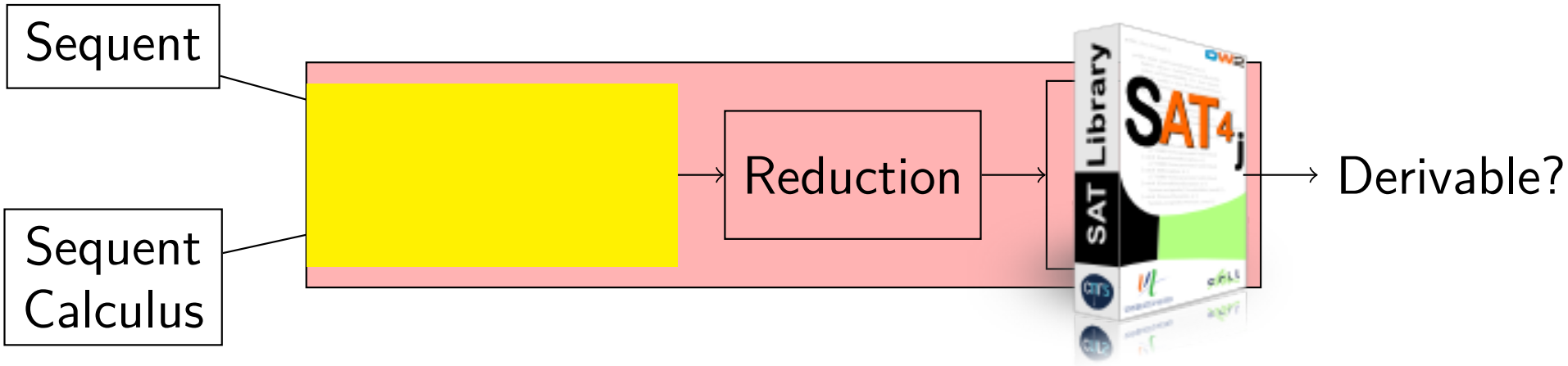
Theorem

There is a polynomial reduction from the derivability problem of any such calculus to (the complement of) SAT.

corollary

The derivability problem for every such calculus is in co-NP.

Tool for Non-classical Logics



Semantics for Pure Calculi

valuations

A valuation is a function $v : WFF \to \{\mathrm{T}, \mathrm{F}\}$

Warning

Valuations are defined over all formulas, not only the atomic ones!

G-legal valuations

A valuation is G-legal if it respects the "semantic reading" of G.

Example (Classical Conjunction)

$$\frac{\Rightarrow A \quad \Rightarrow B}{\Rightarrow A \wedge B} \qquad \frac{A \Rightarrow}{A \wedge B \Rightarrow} \qquad \frac{B \Rightarrow}{A \wedge B \Rightarrow}$$

1. If $v(A) = \mathrm{T}$ and $v(B) = \mathrm{T}$ then $v(A \wedge B) = \mathrm{T}$
2. If $v(A) = \mathrm{F}$ then $v(A \wedge B) = \mathrm{F}$
3. If $v(B) = \mathrm{F}$ then $v(A \wedge B) = \mathrm{F}$

Semantics for Pure Calculi

valuations

A valuation is a function $v : WFF \to \{\mathrm{T}, \mathrm{F}\}$

Warning

Valuations are defined over all formulas, not only the atomic ones!

G-legal valuations

A valuation is G-legal if it respects the "semantic reading" of G.

Example (Sequent Calculus for C_1)

$$\frac{A \Rightarrow}{\Rightarrow \neg A} \qquad \frac{A \Rightarrow}{\neg\neg A \Rightarrow} \qquad \frac{\neg A \Rightarrow \quad \neg B \Rightarrow}{\neg(A \wedge B) \Rightarrow}$$

1. If $v(A) = \mathrm{F}$ then $v(\neg A) = \mathrm{T}$
2. If $v(A) = \mathrm{F}$ then $v(\neg\neg A) = \mathrm{F}$
3. If $v(\neg A) = \mathrm{F}$ and $v(\neg B) = \mathrm{F}$ then $v(\neg(A \wedge B)) = \mathrm{F}$

This semantics is non-deterministic.

Semantics for Pure Calculi

Soundness and Completeness [Béziau '01]

s is provable in G

$$\Longleftrightarrow$$

s is satisfied by every G-legal valuation

Semantics for Pure Calculi

Soundness and Completeness

s is provable in G using $\mathcal{F} \subseteq WFF$

$\Longleftrightarrow$

s is satisfied by every G-legal valuation with domain $\mathcal{F}$

Semantics for Pure Calculi

Soundness and Completeness

s is provable in G using $sub(s)$

$$\Longleftrightarrow$$

s is satisfied by every G-legal valuation with domain $sub(s)$

Semantics for Pure Calculi

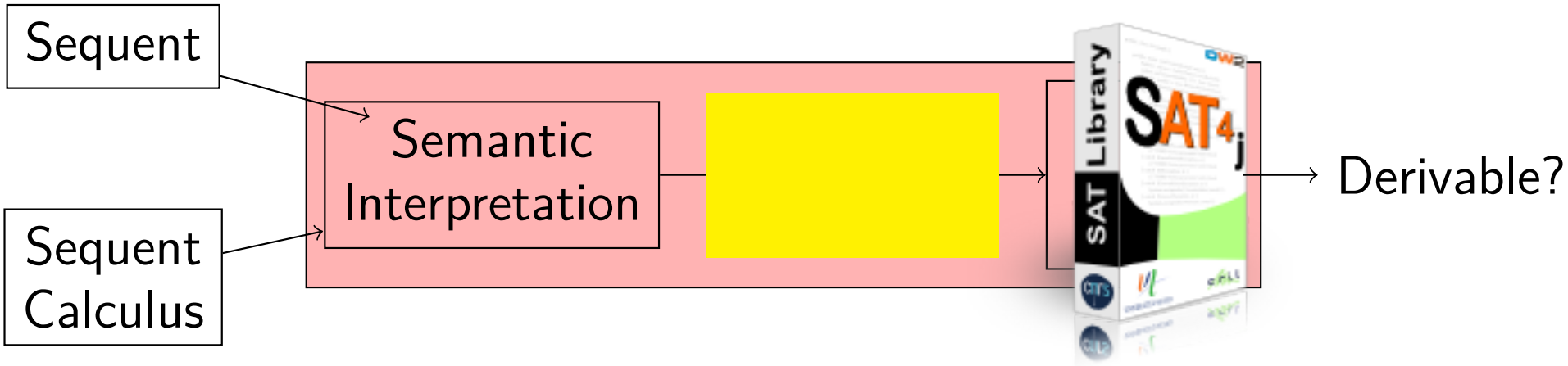
Soundness and Completeness

s is provable in G

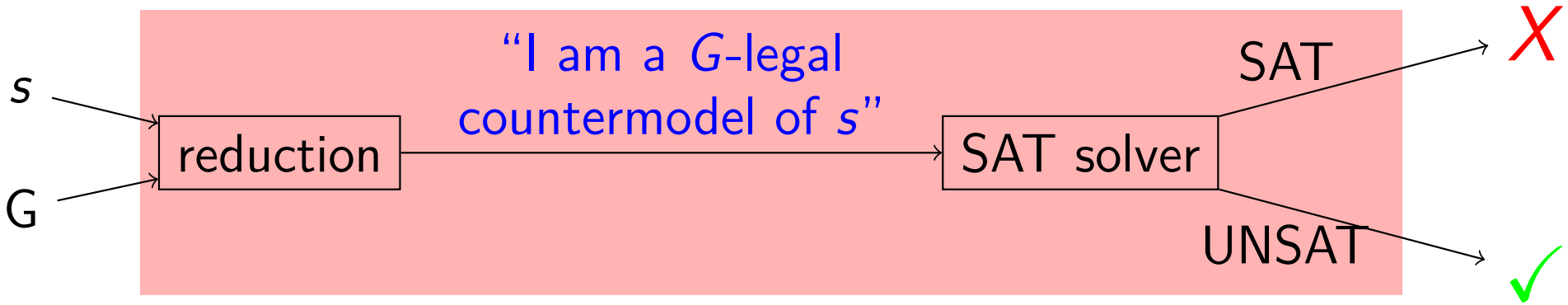
$$\Longleftrightarrow$$

s is satisfied by every G-legal valuation with domain $sub(s)$

Tool for Non-classical Logics



Reduction to SAT



Theorem

- Correctness: s is provable in G iff the generated instance is UNSAT.
- Complexity:
  - Translating: $O(n^k)$; k depends on G ("usually" $k = 1$)
  - Solving: Exp in the worse case. Linear for "Horn calculi"

- In the presence of Next operators, we use Kripke models
- Correctness is more challenging

Propositional logic example

The clauses which define the semantics of propositional logic provide instructive examples of the resolution rule. Here if x and y name propositions x* and y* respectively then

x & y	names the proposition	x* and y*
x ∨ y		x* or y*
x ⊃ y		if x* then y*
x⇔y		x* if and only if y*
¬ x		it is not the case that x*.

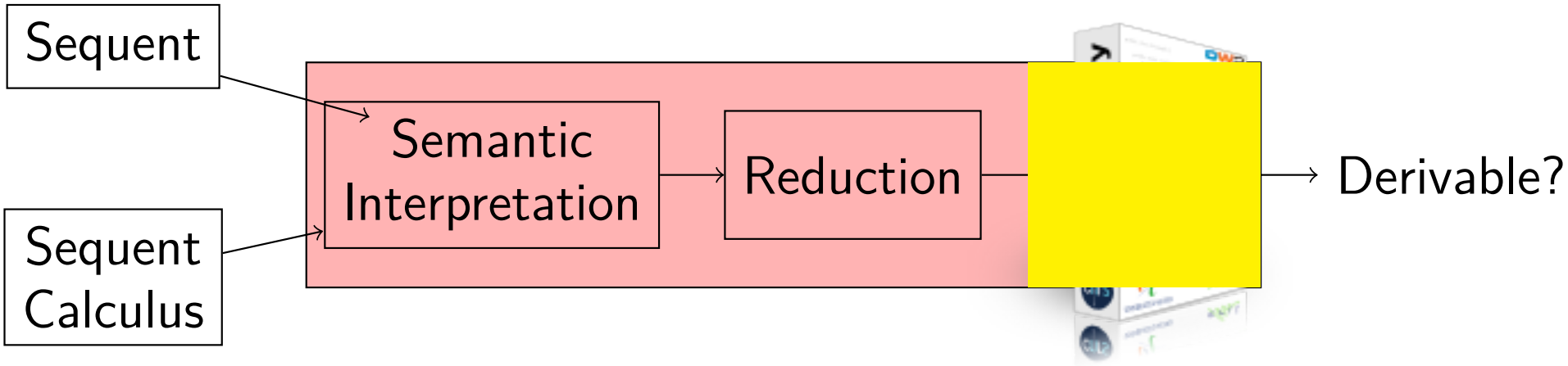
where &, ∨, ⊃, ⇔ and ¬ are infix function symbols. Read True(x) as stating that x is true. The following set of clauses cannot be reexpressed as Horn clauses by renaming predicate symbols.

```
T1    True(x&y) <- True(x), True(y)
T2    True(x)   <- True(x&y)
T3    True(y)   <- True(x&y)
T4    True(x∨y) <- True(x)
T5    True(x∨y) <- True(y)
T6    True(x), True(y)  <- True(x∨y)
T7    True(x⊃y),True(x) <-
T8    True(x⊃y) <- True(y)
T9    True(y)   <- True(x), True(x⊃y)
T10   True(x⇔y) <- True(x y), True(y⊃
T11   True(x⊃y) <- True(x⇔y)
T12   True(y⊃x) <- True(x⇔y)
T13   True(¬x), True(x) <-
T14   <- True(¬x), True(x)
```

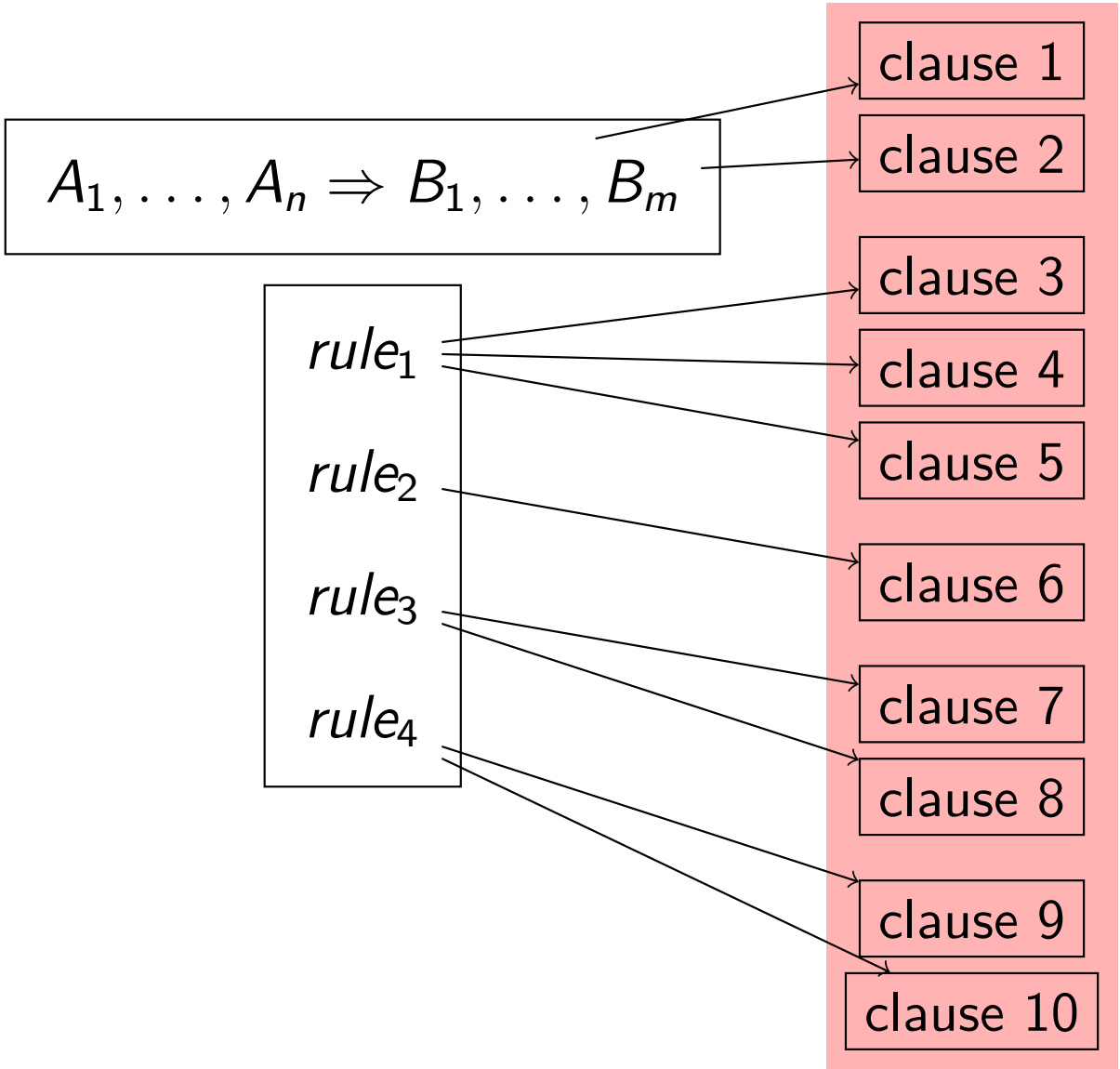
Logic for Problem Solving

Robert Kowalski
Imperial College of Science and Technology
University of London

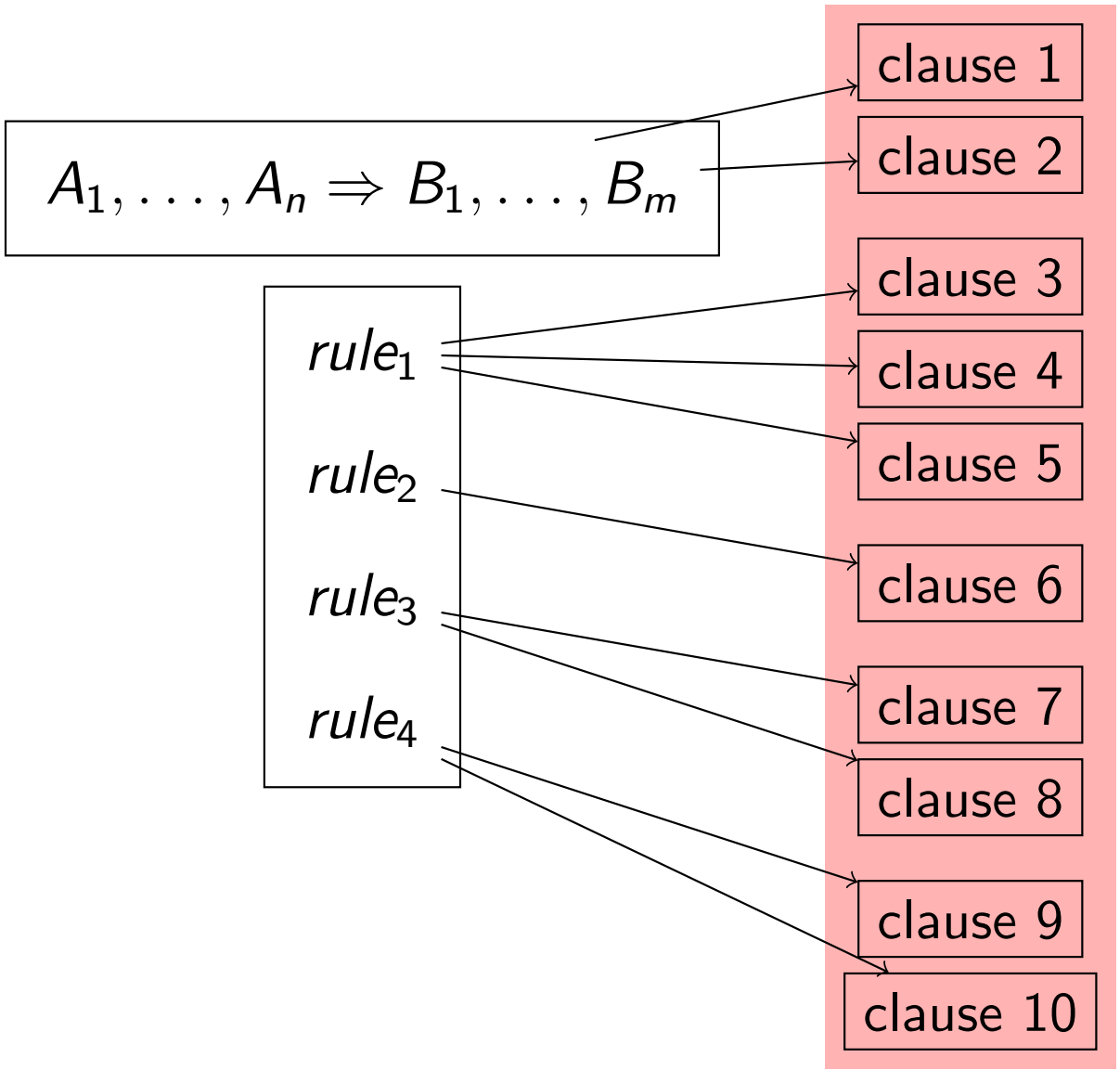
Tool for Non-classical Logics



Reduction


$A_1, \ldots, A_n \Rightarrow B_1, \ldots, B_m$
$rule_1$
$rule_2$
$rule_3$
$rule_4$
clause 1
clause 2
clause 3
clause 4
clause 5
clause 6
clause 7
clause 8
clause 9
clause 10

Reduction

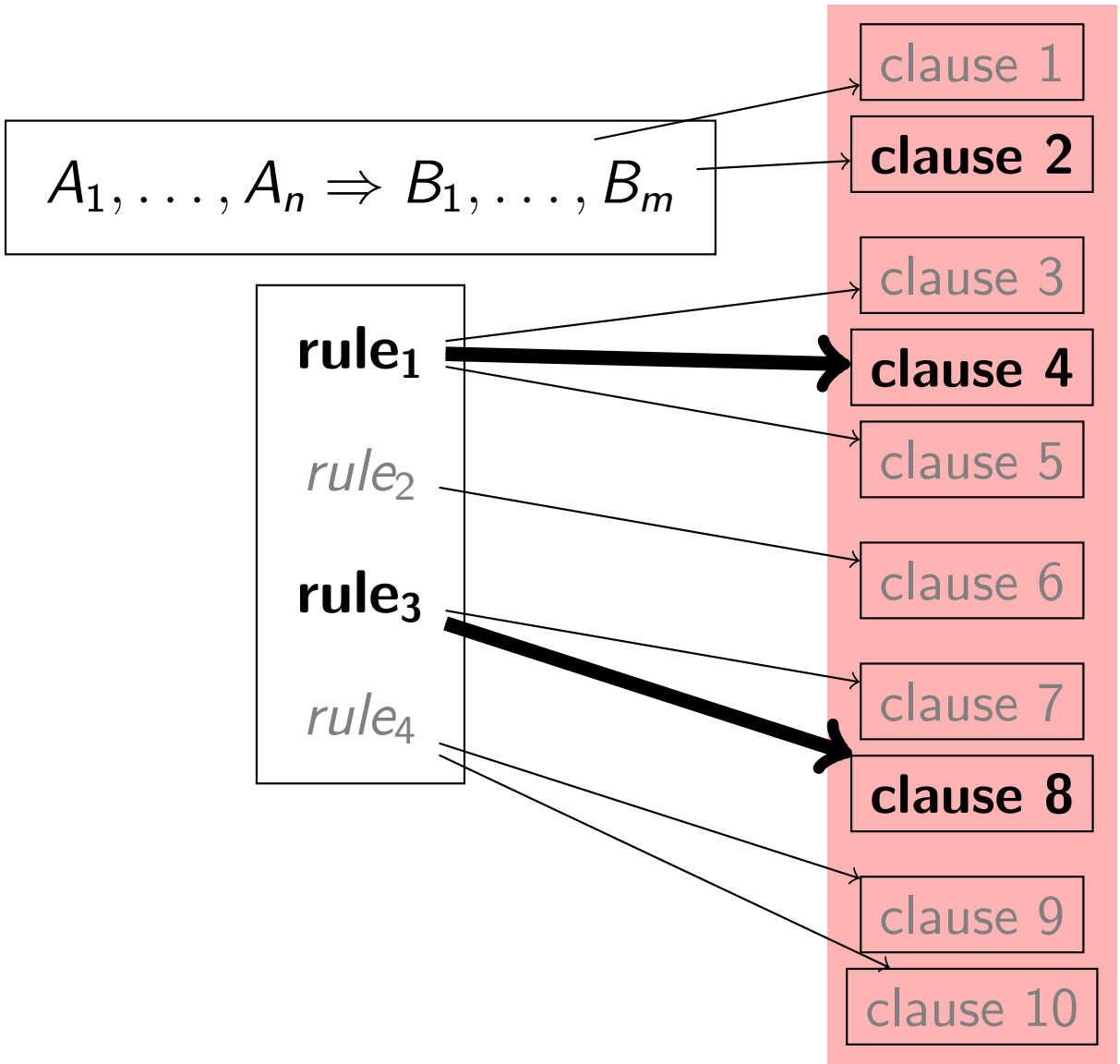


SAT

SAT assignment

$A_1 = \mathit{false}, A_2 = \mathit{true}, \ldots$

Reduction



UNSAT

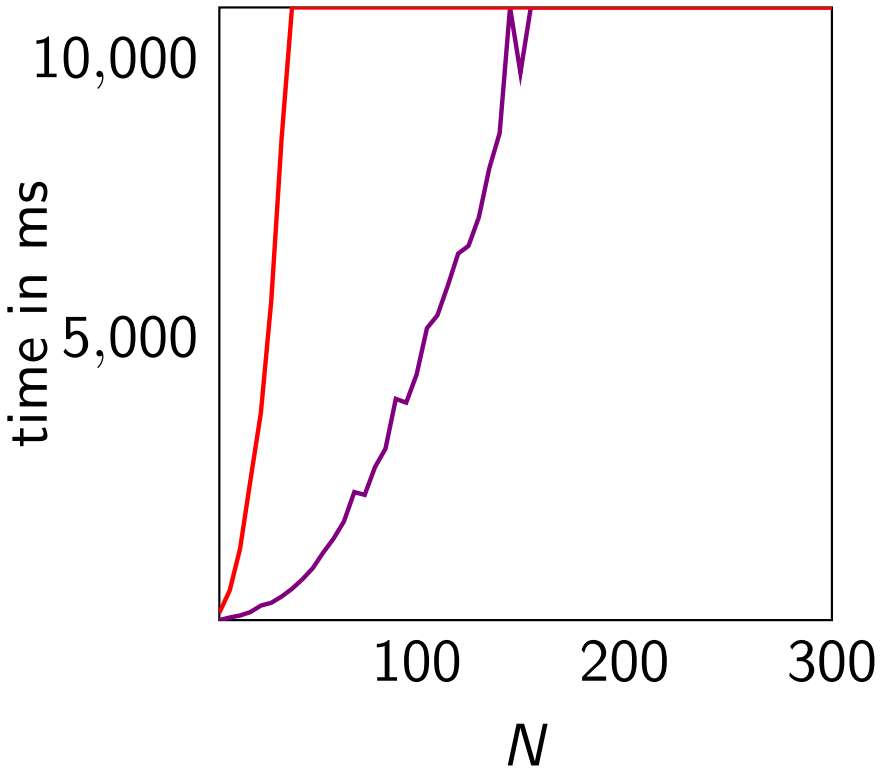
UNSAT core

$2, 4, 8$

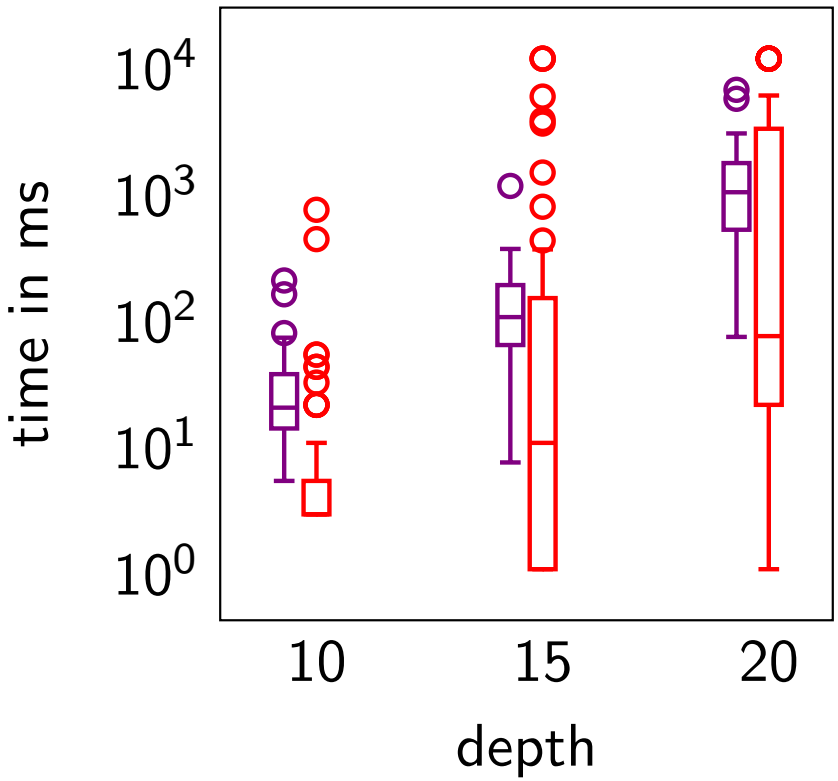
Evaluation

- Gen2sat vs. MetTeL
- input: The calculus for Łukasiewicz 3-valued logic
- Structured Problems for Łukasiewicz infinite-valued logic [Rothenberg'07]
- Random Problems generated by MetTeL

Structured Problems



Random Problems



An Idea: Logic Education

Motivation:

- Gen2sat can be useful for teaching sequent calculi
- Students can focus solely on the logical aspects
- Heuristics and search are left for the SAT solver

Preliminary Pilot:

- 13 logic students were given a bonus assignment:
  present a minimal test plan with maximal coverage for Gen2sat
- They all got 70%-85% coverage
- Some used 0-ary and 3-ary connectives.
- Some found (intentionally planted) bugs
- Feedback from students was encouraging
  - "it helped me see the variety of different connectives and rules"
  - "for me thinking of the extreme cases was really illuminating"
  - "I wish all of the course assignments were more of this type"
  - ...

Conclusion

We have seen:

- A generic tool for deciding derivability in analytic pure (and some impure) sequent calculi
- The actual search is done by a SAT-solver
- Based on a semantic interpretation

Future work:

- Applications
- Support more logics and calculi
- Automatically detect analyticity (when possible)
- Integrate with a theorem prover

Conclusion

We have seen:

- A generic tool for deciding derivability in analytic pure (and some impure) sequent calculi
- The actual search is done by a SAT-solver
- Based on a semantic interpretation

Future work:

- Applications
- Support more logics and calculi
- Automatically detect analyticity (when possible)
- Integrate with a theorem prover

Thank you!